<sol_start id=1>
1. $\sqrt{3}, 2\sqrt{3}, 3\sqrt{2},$ ……

$a = \sqrt{3}$

$d = 2\sqrt{3} - \sqrt{3}$

$= \sqrt{3}$

$a_n = a + (n - 1)d$

$a_{100} = \sqrt{3} + (100 - 1)\sqrt{3}$

$= \sqrt{3} + 99\sqrt{3}$

$= 100\sqrt{3}$

<sol_end>

<sol_start id=2>
2. A circle can have an infinite number of tangents
<sol_end>

<sol_start id=3>
3. n^{th} – term of AP = 8n – 5

a = 8(1) – 5

= 3

Last term in 28^{th} term

l = 8(28) – 5

= 219

$S_n = \frac{n}{2}(a + l)$

$S_{28} = \frac{28}{2}(3 + 219)$

= 3108

<sol_end>

<sol_start id=4>
4. OP = OQ (Radii of same circle)

$\therefore \Delta POQ$ is an isoceles triangle

$\angle OPQ = OQP$

$OP \perp PT$

$\angle OPT = 90°$

$\angle OPQ + QPT = 90$

$55° + \angle TPQ = 90°$

$\angle TPQ = 90° - 55°$

$= 35°$

<sol_end>

<sol_start id=5>

5. AP : 3, 15, 27, 39,……

$a = 3, \quad d = 15 - 3$

$\quad\quad\quad = 12$

$a_n = a + (n - 1)d$

$n = 54$

$n_{54} = 3 + (54 - 1)12$

$\quad = 639$

Term which is 132 more than 54^{th} term is

$639 + 132 = 771$

$a_n = 771$

$771 = a + (n - 1)d$

$771 = 3 + (n - 1)12$

$n = 65$

<sol_end>

<sol_start id=6>

6. PR = 12 cm, OP = 5 cm and OQ = 4 cm

$OP = OQ + OP$

$OP = 4 + 5$

$\quad = 9$ cm

$RO^2 = PR^2 + OP^2$

$RO^2 = 12^2 + 5^2$

$RO = 15$ cm

$RQ^2 = RO^2 - OQ^2$

$RQ^2 = 15^2 - 42$

$RQ = \sqrt{209}$ cm

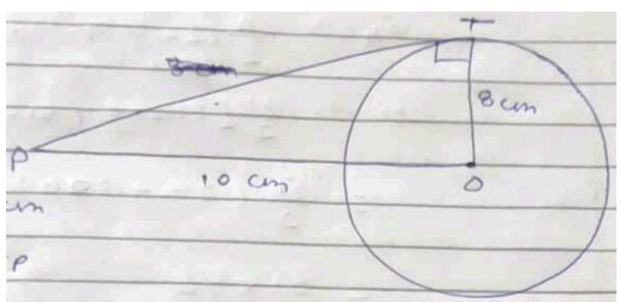
$OP = 5$ cm

$RO^2 = PR^2 + OP^2$

$RO^2 = 15^2$
$RO = 13$ cm
$RQ^2 = RO^2 - OQ^2$
$RQ^2 = 13^2 - 4^2$
$RQ = \sqrt{153}$ cm
<sol_end>

<sol_start id=7>
7. Lowest 3 digit number = 87×2
$= 174$
Highest 3 digit number = 87×11
$= 957$
$l = a + (n - 1)d$
$957 = 174 + (n - 1)87$
$n = 10$
<sol_end>

<sol_start id=8>
8. $OP = 10$ cm
$OT = 8$ cm
$OT \perp PT$
Radius = 8 cm
In ΔOTP
Using Pythagoras Theorem



$OP^2 = OT^2 + PT^2$
$10^2 = 8^2 + PT^2$
$PT^2 = 10^2 - 8^2$
$PT^2 = 100 - 364$
$= 36$
$PT = 6$ cm
Length of tangent = 6 cm
<sol_end>

<sol_start id=9>
9. $a_2 = 14$ $a_3 = 18$
$a + d = 14$ …(1) $a + 2d = 18$ …(2)

$$\begin{array}{l} a + d = 14 \\ \underline{\cancel{a} + 2d = 18} \\ \quad -d = 4 \end{array}$$

d = 4

Putting value of d in eq (1)

a + 4 = 14

a = 10

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_{51} = \frac{51}{2}[2(10) + (51-1)4]$$

$$= \frac{51}{2}(20 + 200)$$

= 51 × 110

S_{51} = 5610

<sol_end>

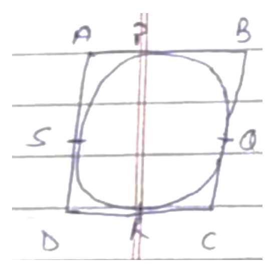
<sol_start id=10>

10. ABCD is a parallelogram circumscribed in circle

AB = CD ...(1)

BC = AD ...(2)

$$\left.\begin{aligned} DR &= DS \\ CR &= CQ \\ BP &= BQ \\ AP &= AS \end{aligned}\right\}$$ (tangents on the circle from the point are same)



Adding all these equations we get

DR + CR + BP + AP = DS + CQ + BQ + AS

CD + AB = AD + BC

Putting the value of eq (1) and (2)

2AB = 2BC

AB = BC ...(3)

From eq (1) (2) and (3) we get

AB = BC = CD = DA

∴ ABCD is a Rhombus

<sol_end>